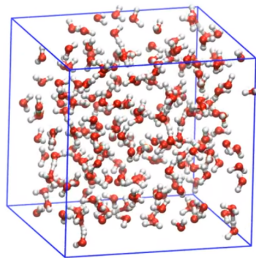


Unit - 03

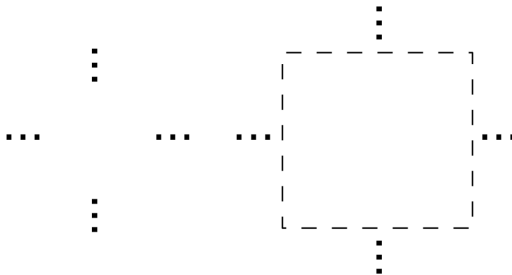
Introduction to Molecular Dynamics



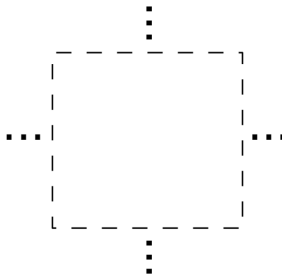
$$\mathbf{F}_i = m_i \mathbf{a}_i \quad i = 1, 2, 3, \dots, N$$

$$\mathbf{F}_i = -\nabla U(\mathbf{r}_i)$$

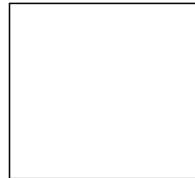
Boundary Conditions



Free space



Periodic



Walls

Program Structure

- ① Parameters and initial conditions;
- ② Forces calculation;
- ③ Integration of motion equations;
- ④ Come back to 2 as many times as wanted.

Parmateres and Initial Conditions

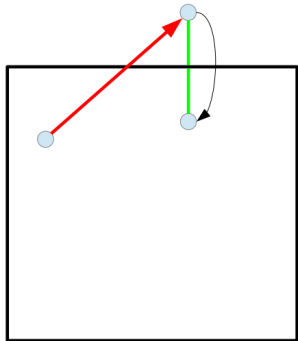
- Cell size;
- Number of particles;
- Time increment;
- Number of time steps;
- Temperature;
- Initial positions and velocities;
- Interaction potential;
- Interaction cutoff.

$$U(\mathbf{r}) = 0$$

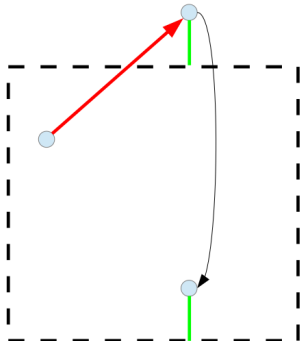
$$\mathbf{r}_{i+1} = \mathbf{r}_i + \mathbf{v}_i dt$$

$$\sum |\mathbf{v}|^2 = T \quad dt \leq 0.01/\sqrt{T}$$

Box limit corrections



$$v_y \rightarrow -v_y$$



$$v_y \rightarrow v_y$$

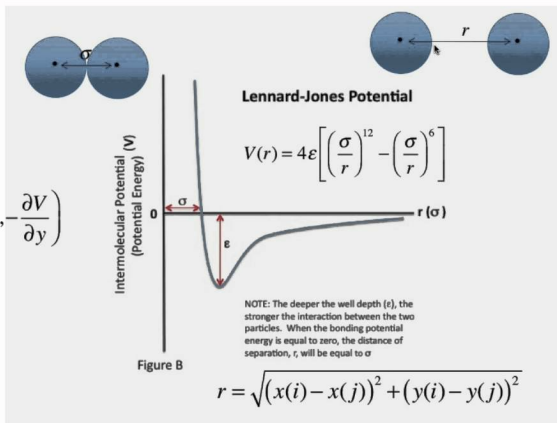
- .xyz format

$$U(r) = q \left\{ \left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right\}$$

$$F(\mathbf{r}) = 12q \left\{ \frac{\sigma^{12}}{r^{14}} - \frac{1}{2} \frac{\sigma^6}{r^8} \right\} \mathbf{r}$$

Lennard-Jones Potential

$$\mathbf{F} = \left(-\frac{\partial V}{\partial x}, -\frac{\partial V}{\partial y} \right)$$



http://chemwiki.ucdavis.edu/Core/Physical_Chemistry/Physical_Properties_of_Matter/Atomic_and_Molecular_Properties/Intermolecular_Forces/Specific_Interactions/Lennard-Jones_Potential

Let us use

$$q = 1 \quad \sigma = 2^{-1/6}$$

$$F(\mathbf{r}) = 3 \left\{ \frac{1}{r^{14}} - \frac{1}{r^8} \right\} \mathbf{r}$$

Let us use

$$q = 1 \quad \sigma = 2^{-1/6}$$

$$F(\mathbf{r}) = 3 \left\{ \frac{1}{r^{14}} - \frac{1}{r^8} \right\} \mathbf{r}$$

→ Superposition Principle

Time increment

$$dt \leq 0.01/\sqrt{T}$$

Verlet integration

$$\mathbf{r}_{i+1} = \frac{\mathbf{F}_i}{m_i}(dt)^2 + 2\mathbf{r}_i - \mathbf{r}_{i-1}$$

$$\mathbf{r}_2 = \mathbf{r}_1 + \mathbf{v}_1 dt + \frac{1}{2} \frac{\mathbf{F}_i}{m_i} (dt)^2$$